Lecture 11 Compendium: GIS for Sustainable Crop Production

Module 3: Technologies of GIS in Crop Protection

Dr. Kadeghe Fue
PhD Agroecology

Compendium for Advanced Study

Contents

1 Introduction: The Sustainability Paradigm **2**

2 Land Suitability Analysis (LSA) **2**
2.1 The FAO Framework (1976) 2
2.2 Multi-Criteria Evaluation (MCE) 2
2.2.1 The Analytic Hierarchy Process (AHP) 2
2.3 Python: Calculating AHP Weights 3

3 Soil Erosion Modeling: RUSLE **3**

4 Precision Conservation **4**
4.1 Topographic Wetness Index (TWI) 4
4.2 Variable Width Buffers 4

5 Agro-Ecological Zoning (AEZ) **4**
5.1 Key Parameters 4

6 Social GIS: Conflict Resolution **4**
6.1 Participatory GIS (PGIS) 4
6.2 Conflict Risk Mapping 4

7 Review Questions & Research Assignment **5**

1 Introduction: The Sustainability Paradigm

Sustainability in agriculture is the optimization of land capability against land use intensity. It is not about "maximum yield at any cost," but "optimal yield within ecological boundaries."

GIS provides the quantitative framework to measure these boundaries. We move from descriptive mapping to **Prescriptive Modeling**.

PhD Research Context

In Tanzania, sustainability is often threatened by: 1. **Soil Degradation:** Erosion due to farming on steep slopes (e.g., Uluguru Mountains). 2. **Land Use Conflict:** Pastoralist vs. Farmer clashes (e.g., Kilosa District). Your research must use GIS to provide evidence-based solutions to these problems.

2 Land Suitability Analysis (LSA)

LSA is the process of determining the fitness of a given tract of land for a defined use (e.g., "Irrigated Rice").

2.1 The FAO Framework (1976)

Standard classification for suitability:

- **S1 (Highly Suitable):** No significant limitations.
- **S2 (Moderately Suitable):** Limitations reduce productivity or increase inputs.
- **S3 (Marginally Suitable):** Inputs are marginally justified.
- **N (Not Suitable):** Physically or economically impossible.

2.2 Multi-Criteria Evaluation (MCE)

Land suitability is multivariate. We must combine different layers (Soil, Climate, Topography) that have different units (pH vs Degrees vs mm).

Standardization: Converting all raw values into a common score (0 to 1). **Weighting:** Assigning importance to each factor.

2.2.1 The Analytic Hierarchy Process (AHP)

AHP is the PhD-standard for determining weights mathematically, reducing subjective bias. We use a **Pairwise Comparison Matrix**.

$$A = \begin{bmatrix} 1 & a_{12} & \dots & a_{1n} \\ 1/a_{12} & 1 & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 1/a_{1n} & 1/a_{2n} & \dots & 1 \end{bmatrix}$$

Where a_{ij} represents how much more important Factor i is compared to Factor j.

- 1 = Equal Importance
- 3 = Moderate Importance
- 9 = Extreme Importance

The weights (w) are the **Principal Eigenvector** of matrix A. We calculate the **Consistency Ratio (CR)** to ensure the expert's judgments are logical. If $CR > 0.1$, the matrix must be re-evaluated.

2.3 Python: Calculating AHP Weights

A snippet to calculate weights from a comparison matrix.

```python
import numpy as np

def calculate_ahp_weights(matrix):
    # 1. Normalize the matrix (column-wise)
    column_sums = matrix.sum(axis=0)
    normalized_matrix = matrix / column_sums

    # 2. Calculate row averages (The Weights)
    weights = normalized_matrix.mean(axis=1)

    return weights

# Example: Suitability for Rice
# Factors: [Rainfall, Slope, Soil_Texture]
# Logic: Rainfall is moderately more important than Slope (3)
#        Rainfall is strongly more important than Texture (5)
comparison_matrix = np.array([
    [1,   3,   5],  # Rainfall
    [1/3, 1,   2],  # Slope
    [1/5, 1/2, 1]   # Texture
])

weights = calculate_ahp_weights(comparison_matrix)

print("--- Feature Weights ---")
print(f"Rainfall: {weights[0]:.3f}")
print(f"Slope:    {weights[1]:.3f}")
print(f"Texture:  {weights[2]:.3f}")
```

Listing 1: Calculating AHP Weights

3 Soil Erosion Modeling: RUSLE

Soil loss is a physical process that can be modeled spatially. The **Revised Universal Soil Loss Equation (RUSLE)** is the global standard.

$$A = R \times K \times LS \times C \times P \tag{1}$$

Where:

- **A:** Annual soil loss (tons/ha/year).
- **R (Rainfall Erosivity):** Derived from intensity-duration data (Meteorology).
- **K (Soil Erodibility):** Function of texture and organic matter (Soil Map).
- **LS (Slope Length & Steepness):** Derived from the DEM (Digital Elevation Model). This is the GIS-heavy component.
- **C (Cover Management):** Derived from NDVI. Bare soil has high C; Forest has low C.
- **P (Support Practice):** Effect of terracing or contour farming.

Research Application: A PhD thesis might involve simulating how changing P (introducing terraces) reduces A (Soil Loss) in the Usambara mountains using GIS scenario modeling.

4 Precision Conservation

Instead of applying conservation practices everywhere (expensive), we use GIS to identify **Critical Source Areas (CSAs)**the specific 10% of the land contributing 80% of the sediment load.

4.1 Topographic Wetness Index (TWI)

Predicts where water accumulates.

$$TWI = \ln\left(\frac{\alpha}{\tan\beta}\right)$$

Where α is the upslope contributing area and β is the local slope. High TWI areas are prone to saturation and runoff.

4.2 Variable Width Buffers

Standard policy requires a 60m buffer along all rivers. Precision Conservation suggests **Variable Width Buffers**wider buffers where runoff flow paths (high TWI) intersect the river, and narrower buffers elsewhere. This saves land for farmers while maximizing environmental protection.

5 Agro-Ecological Zoning (AEZ)

AEZ is land resource mapping at a regional/national scale (FAO). It divides a country into zones with similar production potential.

5.1 Key Parameters

1. **LGP (Length of Growing Period):** Number of days where Precipitation + Soil Moisture Storage > 0.5 * Potential Evapotranspiration (PET).
2. **Thermal Regime:** Temperature sums (Degree Days) available during the LGP.

GIS Operation: Intersection (Overlay) of Climate Surfaces and Soil Polygons.

6 Social GIS: Conflict Resolution

Sustainability includes social stability. In Tanzania, land use conflict is a major issue.

6.1 Participatory GIS (PGIS)

PGIS combines expert spatial data with local indigenous knowledge. * **Method:** Community mapping workshops where elders draw traditional cattle corridors and water points on satellite image printouts. These are then digitized. * **Application:** Overlaying "Official" land use plans with "Traditional" usage to identify conflict hotspots (intersections).

6.2 Conflict Risk Mapping

Using GIS to model the probability of conflict. * **Variables:** Proximity to water points (dry season), Crop density, Cattle density tracks. * **Output:** A heatmap identifying areas where farmers and pastoralists are forced into competition.

7 Review Questions & Research Assignment

1. **Modeling (RUSLE):** You are modeling erosion in a watershed. If you resample your DEM from 30m resolution to 90m resolution, what happens to the calculated LS factor? Does it over-estimate or under-estimate erosion? Why?

2. **Decision Science (AHP):** Why is it necessary to calculate the "Consistency Ratio" in AHP? What does it imply about human decision-making?

3. **Policy:** Critically analyze the "60m River Buffer" law in Tanzania from a Precision Conservation perspective. How would you use GIS to propose a more efficient alternative policy?

References

1. FAO. (1976). *A Framework for Land Evaluation*. Soils Bulletin 32.

2. Renard, K. G., et al. (1997). *Predicting soil erosion by water: a guide to conservation planning with the Revised Universal Soil Loss Equation (RUSLE)*. USDA.

3. Saaty, T. L. (1980). *The Analytic Hierarchy Process*. McGraw-Hill.